Oral Exam: Bayesian Statistics and Data Analysis (2ST128)

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Oral Exam Instructions

The oral exam will start at the scheduled time, and you need to be there when it begins. We, as examiners, will then randomize the order of students. We will continuously inform you of the next two students in the queue so you can prepare and wait for your turn. This means that if you are not next in line, you can go to the toilet, etc., and do not need to be immediately prepared. If you do not show up when called, you will fail that oral exam opportunity.

During the exam, you will get one random question for each assignment covered by that oral exam. This means that an oral exam covering several assignments can contain several questions, one from each relevant assignment section. The question will be drawn from the list of questions below, and you will answer it orally or with the help of a whiteboard and pen to show that you know the material. If you fail the first question for an assignment, two new random question will be drawn from the same assignment section (however, not the already failed question). If any of the second question also are failed, you will fail the oral exam for that assignment. If you fail the oral exam for an assignment, you cannot get VG for that assignment. Otherwise, you will have passed the oral exam for that assignment. If you have done the reading assignments and the turn-in assignments, this should not be hard. However, I recommend reviewing the questions before the exam to be prepared. Ideally get together in groups and ask the questions to each other.

We can also decide to ask an additional question about the already turned-in assignments if we find the assignments to be strange or suspect that it might have been generated without knowing what was done. Note that you should be able to explain what you did in all turned-in assignments.

There will be three ordinary oral exams during the course, covering assignments 1–3, 4–6, and 7–8, respectively. For each oral exam, there will be the scheduled ordinary opportunity and two additional re-examination opportunities, for a total of three attempts. Later oral exam opportunities will also act as re-examinations for previously failed or missed oral exams covering the same assignments. Then both the reexam and the new exam will be done at the same time. If you fail all three attempts for an oral exam, you will need to retake the course in full next year.

Oral Exam Questions

Assignment 1: Probability, Bayes' Rule, and Basic Computing

1. **Probability mass versus density. Explain the difference between a probability mass function $p_X(x)$ and a probability density function $f_X(x)$. Why can a density be larger than one for a given value of x?**

2. **Likelihood. For a model $p(y \mid \theta)$, what is the likelihood and how is it different from a probability distribution for θ?**

3. **Bayes' rule for a diagnostic test. Write Bayes' rule for $P(D \mid +)$, where D is disease and $+$ is a positive test. Explain why a high test sensitivity does not by itself imply a high posterior probability of disease.**

4. **Product and sum rules. State the product rule and the sum rule for two events A and B. How do they combine to produce Bayes' rule?**

5. **Prior, data, posterior. In one or two equations, explain the Bayesian update from prior to posterior. What is the role of the normalizing constant?**

6. **Beta distribution from mean and variance. Suppose $X \sim \mathrm{Beta}(\alpha, \beta)$. Explain what the parameters α, β and what is the mean and variance of the beta distribution.**

7. **Simulation check. If you sample 1000 values from a Beta distribution and compute the sample mean, why should it be close to the analytical mean? Why will it not match exactly?**

8. **Central probability interval. Define a central 95% probability interval from simulated posterior draws. How would you compute for a parameter θ that is Beta distributed?**

9. **Aleatoric and epistemic uncertainty. Explain aleatoric and epistemic uncertainty using a simple data analysis example, e.g. using linear regression as an example.**

10. **Conditional probability. Suppose that if $\theta = 1$, then y has a normal distribution with mean 1 and standard deviation σ, and if $\theta = 2$, then y has a normal distribution with mean 2 and standard deviation σ. Also, suppose $\Pr(\theta = 1) = 0.5$ and $\Pr(\theta = 2) = 0.5$. For $\sigma = 2$, write the formula for the marginal probability density for y.**

Assignment 2: One-Parameter Models and Conjugacy

1. **Beta-binomial posterior. Let $y \mid \pi \sim \text{Binomial}(n, \pi)$ and $\pi \sim \text{Beta}(\alpha, \beta)$. Derive the posterior distribution up to proportionality. What happens in the special case of a uniform prior?**

2. **Posterior as compromise. In the beta-binomial model, explain how the posterior mean is a compromise between prior information and data.**

3. **Posterior probability from a posterior distribution. How do you compute $P(\pi < 0.2 \mid y)$ in a beta-binomial model? Give both an analytical and simulation answer (i.e. explain what you would do).**

4. **Prior sensitivity. What is prior sensitivity analysis and what would you look for in the algae beta-binomial exercise?**

5. **Conjugacy. Define conjugacy. Why is it pedagogically useful even if later assignments use simulation and Stan?**

6. **Gamma-Poisson posterior. Let $y_i \mid \lambda \sim \text{Poisson}(\lambda)$ independently and $\lambda \sim \text{Gamma}(\alpha, \beta)$ with rate β. Derive the posterior.**

7. **Posterior predictive distribution. What is the posterior predictive distribution and how is it obtained in the Gamma-Poisson model?**

8. **Prior predictive versus posterior predictive distribution. What is the difference between the prior predictive distribution and the posterior predictive distribution?**

9. **Model checking with predictive simulations. In the warpbreaks Poisson exercise, why do we compare histograms of observed data and posterior predictive simulations?**

10. **Credible interval. What is a 95% credible interval for λ in the Gamma-Poisson model and how should it be interpreted?**

11. **Weakly informative prior. What is a weakly informative prior? Give an example in a one-parameter model.**

Assignment 3: Normal Models and Group Comparisons

1. **Normal model with unknown mean and variance. For observations $y_i \sim N(\mu, \sigma^2)$ and the prior $p(\mu, \sigma) \propto 1/\sigma^2$, what is the form of the marginal posterior for μ?**

2. **Posterior interval versus predictive interval. Why is the predictive interval for a future windshield hardness wider than the posterior interval for the mean hardness?**

3. **Posterior predictive density. Write the posterior predictive distribution conceptually for a future observation $\tilde{y}$ in the normal model.**

4. **Difference between two proportions. For two independent binomial groups with beta priors, how can posterior simulation be used to summarize an odds ratio?**

5. **Avoiding null-hypothesis wording. In a Bayesian two-group comparison, why is it often better to report posterior probabilities and intervals than to say whether a point null is rejected?**

6. **Difference between two normal means. Describe a simulation algorithm for the posterior distribution of $\mu_1 - \mu_2$ when each group has a normal model with unknown variance.**

7. **Marginalization. Explain marginalization in the context of obtaining $p(\mu \mid y)$ from $p(\mu, \sigma \mid y)$.**

8. **Conjugate versus noninformative normal priors. What is the difference between using a conjugate normal-inverse-chi-square prior and the noninformative prior in the normal model?**

9. **Continuous posterior and equality. If $\delta = \mu_1 - \mu_2$ has a continuous posterior, what is $P(\delta = 0 \mid y)$? What should we ask instead?**

10. **Communicating uncertainty. Give a good one-sentence Bayesian interpretation of a posterior interval for the odds ratio.**

Assignment 4: Monte Carlo Integration and Importance Sampling

1. **Bioassay model. Write the bioassay logistic regression model used for dose-response data for dose x_i, number of animals n_i, and deaths y_i.**

2. **Bivariate normal prior covariance. If $\alpha \sim N(m_\alpha, s_\alpha^2)$, $\beta \sim N(m_\beta, s_\beta^2)$, and $\text{corr}(\alpha, \beta) = \rho$, write the prior mean vector and covariance matrix.**

3. **Monte Carlo standard error. What is Monte Carlo standard error and why should it affect how many digits are reported?**

4. **Importance sampling identity. State the basic importance sampling identity for estimating $E_p[h(\theta)]$ using draws from a proposal $g(\theta)$.**

5. **Log weights. Why compute importance ratios on the log scale?**

6. **Prior as proposal. In the bioassay importance sampling exercise, what can go wrong if the proposal distribution is the prior?**

7. **Effective sample size for importance sampling. Give the common formula for the effective sample size from normalized importance weights and interpret it.**

8. **Posterior mean from weighted draws. How do you estimate the posterior mean of α from importance sampling draws and normalized weights $\tilde{w}_s$?**

9. **From posterior draws to posterior summaries. Given independent posterior draws of (α, β), how do you estimate posterior means and 90% intervals?**

Assignment 5: Metropolis Algorithm for Bioassay

1. **Metropolis target. In the Metropolis bioassay assignment, what is the target distribution?**

2. **Metropolis acceptance probability. For a symmetric proposal $q(\theta^* \mid \theta) = q(\theta \mid \theta^*)$, write the Metropolis acceptance probability.**

3. **Metropolis-Hastings correction. How does the acceptance probability change for a non-symmetric proposal?**

4. **Choosing proposal scale. What happens if the Metropolis proposal scale is too small? What happens if it is too large?**

5. **Starting values. Why should a report state the starting values or how they were generated for MCMC chains?**

6. **Warm-up. What is warm-up or burn-in in MCMC, and why are warm-up draws not used for posterior summaries?**

7. **Trace plots. What should a good trace plot look like for a scalar parameter? What warning signs should an examiner look for?**

8. **Autocorrelation. Why does autocorrelation in MCMC draws reduce the information in a fixed number of iterations?**

9. **Scatter plot of posterior draws. Why is a scatter plot of (α, β) useful in the bioassay model?**

10. **Detailed balance. State, at a high level, why the Metropolis algorithm has the desired posterior as its stationary distribution.**

Assignment 6: Stan, Diagnostics, and Linear Models

1. **Stan model blocks. Name (at least) three Stan blocks used in a simple model and what each does.**

2. **Bioassay in Stan. Write the key Stan-style likelihood statement for the bioassay model using vector `x`, counts `y`, trials `n`, and parameters `alpha`, `beta`.**

3. **$\widehat{R}$. What does $\widehat{R}$ diagnose and what values are concerning?**

4. **Effective sample size in Stan output. What does effective sample size mean in Stan output, and why can it be much smaller than the number of draws?**

5. **Divergent transitions. What is a divergent transition in HMC/NUTS?**

6. **Generated quantities. Why use Stan's `generated quantities` block for posterior predictive draws?**

7. **Linear Gaussian model. Write a simple Bayesian linear regression model for drowning counts using year as a predictor.**

8. **Prior predictive checks. What is a prior predictive check in a regression model, and why is it useful before fitting the model in Stan?**

9. **Syntax versus statistical errors. In Stan code, what is the difference between (Stan) syntax errors from statistical/modeling errors.**

10. **Posterior predictive for a future year. How would you compute the predictive distribution for drownings in a future year such as 2030?**

Assignment 7: Hierarchical Models

1. **Separate, pooled, and hierarchical models. Explain the difference between separate, pooled, and hierarchical models for the factory data.**

2. **Partial pooling. What is partial pooling and why is it useful with only five observations per machine in the factory example?**

3. **Hierarchical normal model. Write a hierarchical normal model (on μ) for factory measurements y_{ij}.**

4. **Exchangeability. What does exchangeability mean in the hierarchical factory model?**

5. **Prediction for a new machine. How does a hierarchical model predict the mean quality of a seventh, previously unobserved machine?**

6. **Hyperparameters. Interpret θ and τ in the hierarchical normal model for factory measurements. Start by writing down the full hiearchical normal model.**

7. **Funnel geometry. Why can hierarchical models create a funnel-shaped posterior involving μ_j and $\log \tau$?**

8. **Truncated normal priors for scale. Why use a half-normal or otherwise positive prior for σ and τ?**

9. **Comparing predictive distributions. Why can the predictive distribution for machine 6 differ across separate, pooled, and hierarchical models?**

Assignment 8: Model Checking, LOO-CV, and Predictive Accuracy

1. **Posterior predictive checking. What is posterior predictive checking and what question does it answer?**

2. **Predictive accuracy. What is expected log predictive density, and why is it useful for model comparison?**

3. **LOO-CV. Describe leave-one-out cross-validation for Bayesian models.**

4. **Why PSIS?. Why is PSIS-LOO used instead of exact LOO refitting in the assignment?**

5. **Log likelihood matrix. What log likelihood quantity must be saved from Stan to compute LOO, and what is its shape?**

6. **Pareto $\hat{k}$. What does the Pareto $\hat{k}$ diagnostic tell us in PSIS-LOO?**

7. **Model comparison uncertainty. If one model has a slightly higher elpd than another, what else should be considered before declaring it better?**

8. **Effective number of parameters. What is the effective number of parameters in predictive model assessment, conceptually?**

9. **LOO units. Why must the unit left out in cross-validation match the prediction task, especially in hierarchical models?**

10. **Model checking versus model comparison. Explain the difference between checking whether a model is adequate and comparing predictive accuracy between models.**